

Silvanus Brian Mulu

✉ uvb20@txst.edu | ☎ +1 (737) 213 7882 | [in LinkedIn](#) | [🌐 Personal Portfolio](#)

Education

Texas State University

B.A. Economics (Honors) & Minor in Mathematics

• GPA: 4.0

San Marcos, TX

May 2029

Experience

Texas State University, Department of Economics & Finance

Undergraduate Research Assistant

Technologies: R, MATLAB, LaTeX

[\[🔗 Code\]](#) | [\[Work-In-Progress\]](#)

- Building a structural **input-output VAR (SVAR)** in R and MATLAB to trace **oil price shock transmission** through US industry linkages; decomposing **spot vs. speculative shocks** and estimating **impulse response functions** to quantify spillovers onto inflation, output, and the trade balance.

San Marcos, TX

Jan 2026 – Present

National Social Security Fund

Data Analyst Intern

Technologies: Power BI, Excel, Python

- Built **Power BI** dashboards to monitor payment flows and portfolio profitability across the national pension fund portfolio, enabling real time performance tracking
- Performed **pivot table analysis** on financial statements and reports to identify trends in fixed income, equities and real estate assets
- Collaborated with backend engineers on **schema design** and **data validation** rules for the member registration system

Nairobi, Kenya

Mar 2025 – Jun 2025

Projects

Statistical Arbitrage: Pairs Trading in U.S. Oil Equities

[\[🔗 Code\]](#) | [\[📄 Paper\]](#)

Technologies: R, Python, LaTeX (Work-In-Progress)

- Applied cointegration-based pairs trading to U.S. energy equities in **R**, showing $COP - XOM$ fails under correct **Engle-Granger** critical values and that its risk-adjusted performance reflects hedging-driven variance reduction rather than genuine mean reversion.

Newton Forward Method of Interpolation

[\[🔗 Code\]](#) | [\[📄 Paper\]](#)

Technologies: JavaScript, Node.js

- Implemented **Newton's forward-difference interpolation** from scratch using recursive divided differences an exploration on piecewise polynomial fitting up to **30th order**.

Leadership & Extracurriculars

Financial Management Association

Vice President, Quant Finance Cohort

- Lead quantitative markets education and technical workshops for club members while supporting recruitment and professional development.

San Marcos, TX

August 2026 – Present

International Student Organisation

Treasurer, Head of Finance Committee, Board of Director

- Oversee annual budget and financial planning for a 1,500+ member organization, ensuring fiscal accountability and strategic fund allocation.

San Marcos, TX

Sept 2024 – Present

Table Tennis Club

Active Member

- Bronze medallist at a collegiate tournament in eSwatini; 6-time gold medallist at high school tournaments in Kenya.

Mbabane, eSwatini

Jan 2022 – Present

Awards

Shelby Davis Scholar: Full tuition, Texas State

President's List (2X): Perfect GPA

Goldman Sachs: Possibilities Series Participant (2026)

Beta Gamma Sigma: Invited Member, Honor Society (Top 0.5)

MIT Policy Hackathon: Globalist Finalist (2024)

Jump Trading Probability Cup: Top 5% algorithm forecast (2026)

Skills & Interests

- **Certifications:** Securities Industry Essential (SIE), FINRA - in progress
- **Research Interests:** Time Series Econometrics, Energy Economics, Empirical Macroeconomics, Machine Learning, Derivatives & Market Microstructure
- **Programming & Tools:** R, Python, JavaScript, Excel (Advanced), Power BI, Git, GitHub, Linux.
- **Languages:** English (Native), Swahili (Native), Kikamba (Native), Turkish (A2 – Beginner).
- **Interests:** Long-distance running, Poker, Chess, Checkers, Sudoku, Golf, Farming (300 chickens).